Motherson Automotive & Technology engineering
Survery No 172/1-5,274/2,276/7,
Chengalpattu Sriperumbudur Road, Vadgal Village B Vallam Panchayat,,Vadagal,
Kanchipuram, Tamil Nadu, 602105

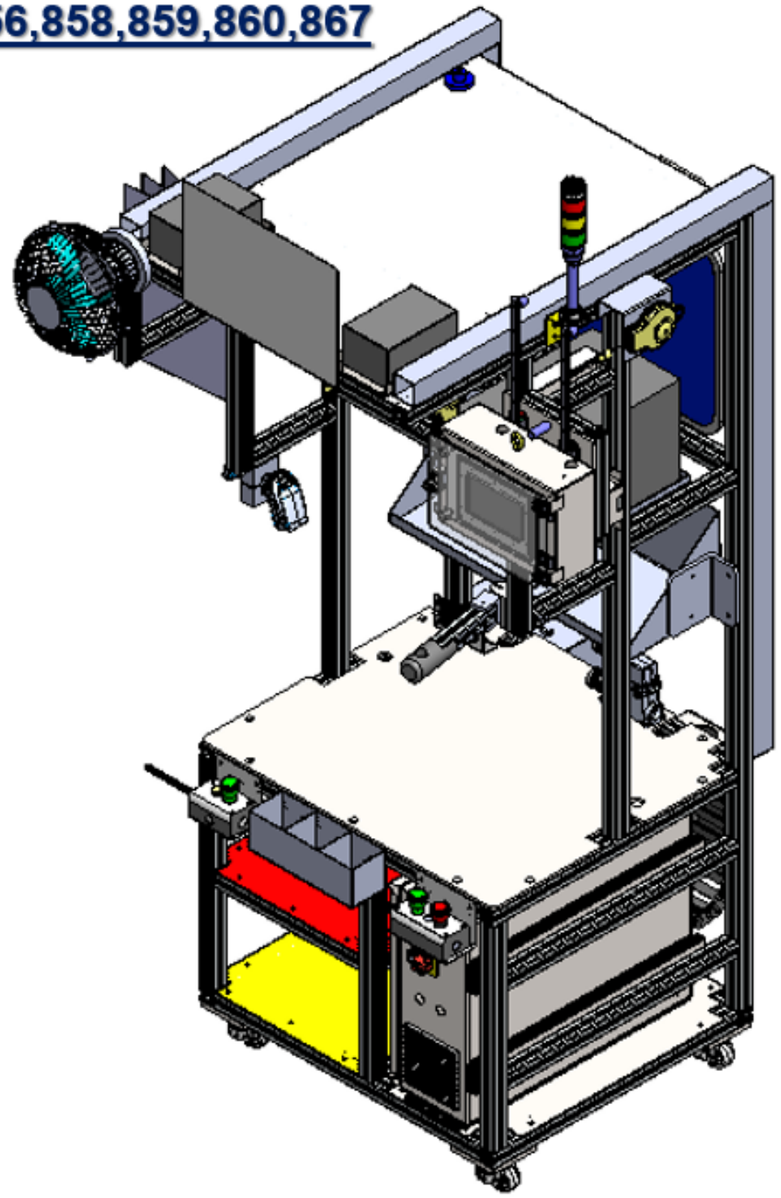
CUSTOMER NAME	MMLI PUNE
PROJECT NAME	S220 HL HV & LV LIGHT STATION-3
PROJECT CODE	FMSPM00856
COMMISSION	ROBIS
PROJECT TYPE	WORK STATION
PLACE OF INSTALLATION	MMLI PUNE
RESPONSIBLE OF PROJECT	ROBIS
REVISION	01(Screw driver wiring updated)
DESIGNER	MANORAJ.S

CREATED ON	10-07-2023	CHECKED BY			
LAST EDITED DATE	11-04-2024	CHECKED ON	29-07-2023	NUMBER OF PAGES	31

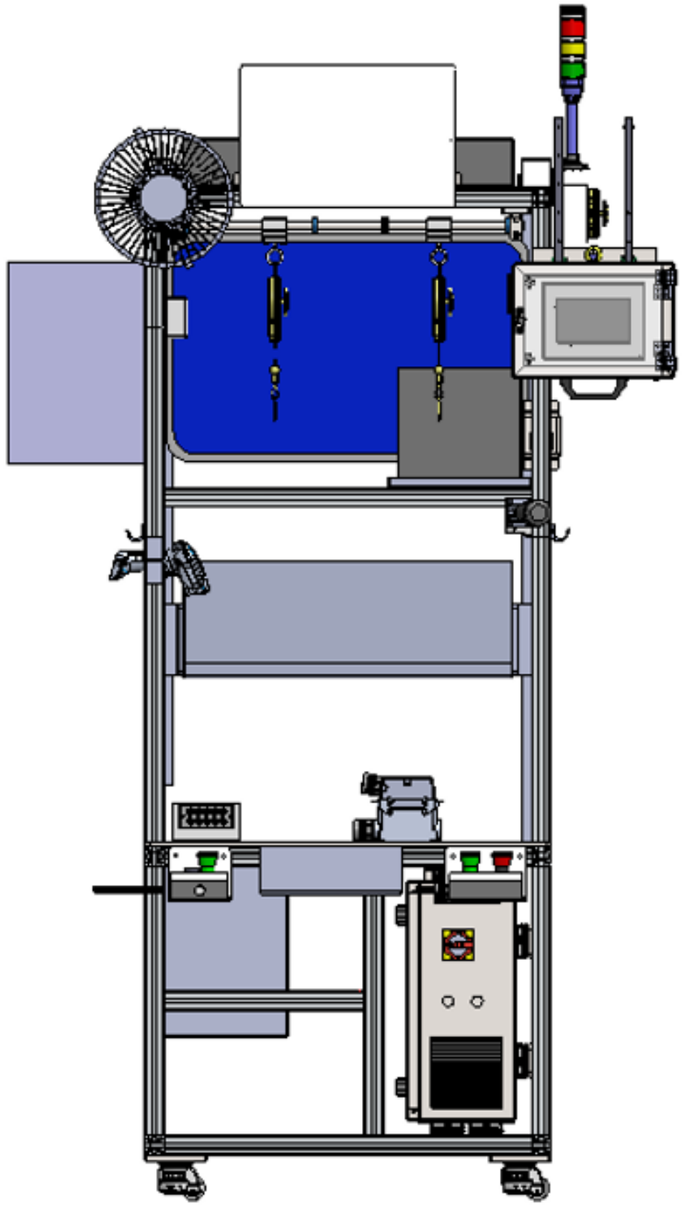
+CP/1

MACHINE OVER VIEW

FMSPM00856,858,859,860,867



ISO VIEW



FRONT VIEW

3

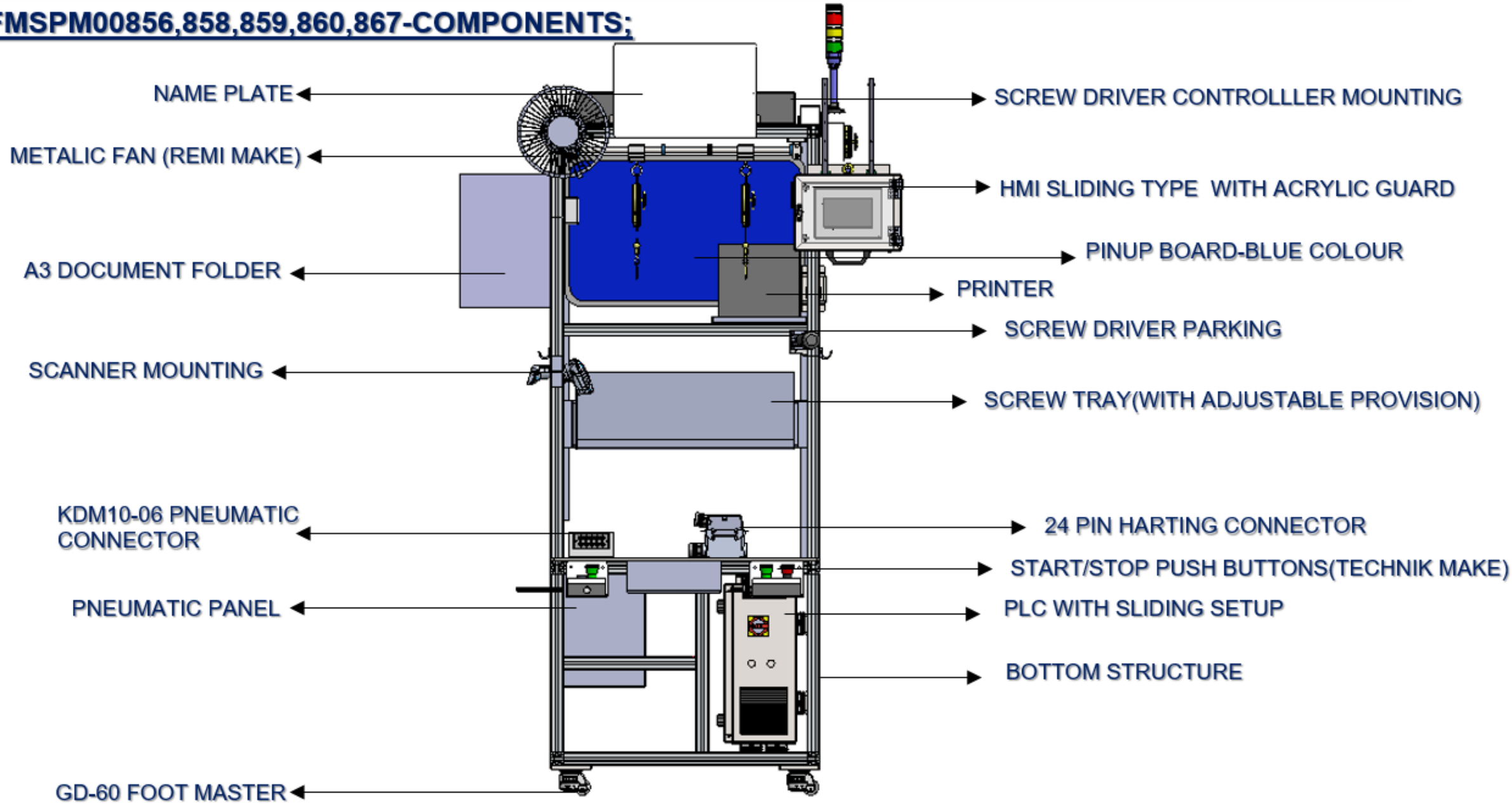
NOTE : PRINTER NOT CONSIDER

REVISION			Date	24-07-2023	ROBIS	Motherson Automotive & Technology engineering	MACHINE OVER VIEW		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 1
Modification	Date	Name	Original		Replacement of	Replaced by			Page 2 / 31

0	1	2	3	4	5	6	7	8	9

MACHINE FRONT VIEW

FMSPM00856,858,859,860,867-COMPONENTS;



NOTE : PRINTER NOT CONSIDER

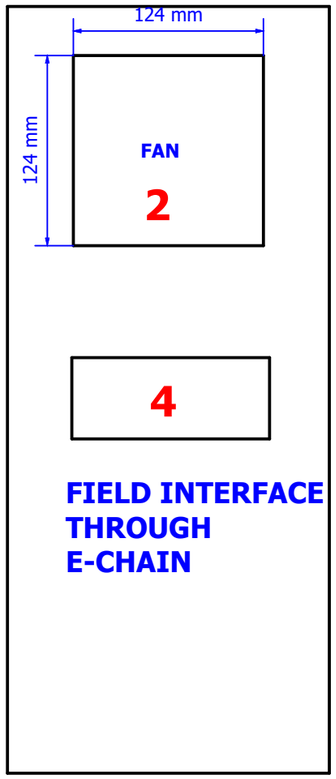
1 3

REVISION			Date	24-07-2023	ROBIS	Motherson Automotive & Technology engineering	MACHINE FRONT VIEW		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 2
Modification	Date	Name	Original		Replacement of	Replaced by			Page 3 / 31

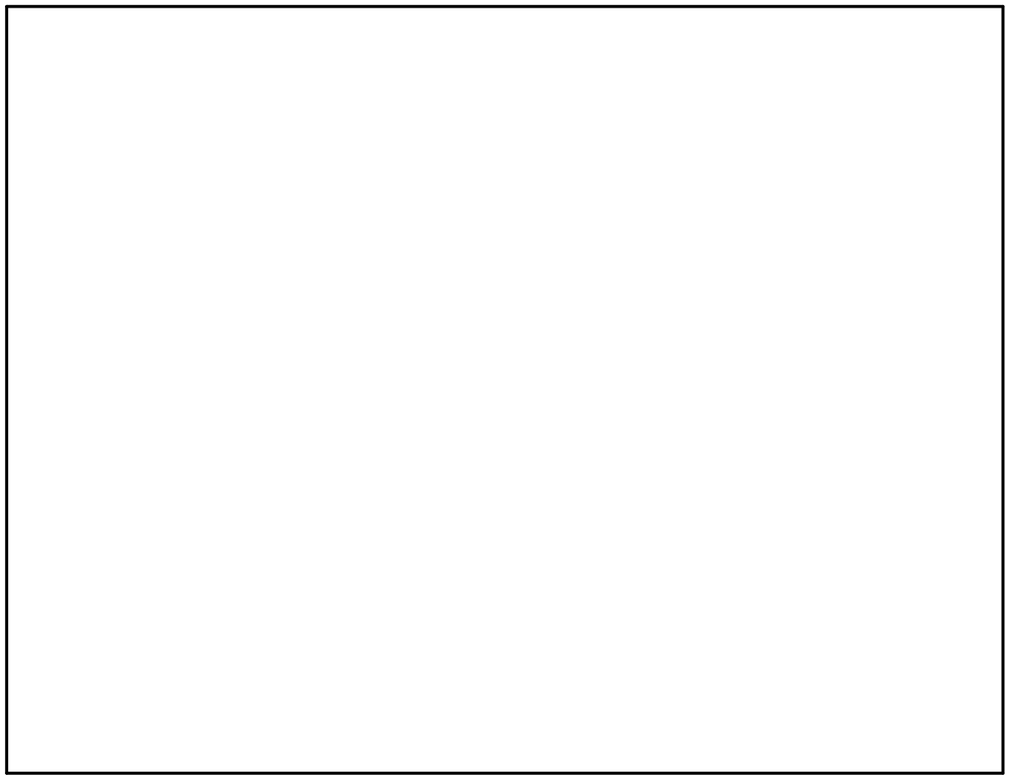
PANEL CUTOUT

CUSTOMIZED PANEL
H 500 X W 650 X D 210

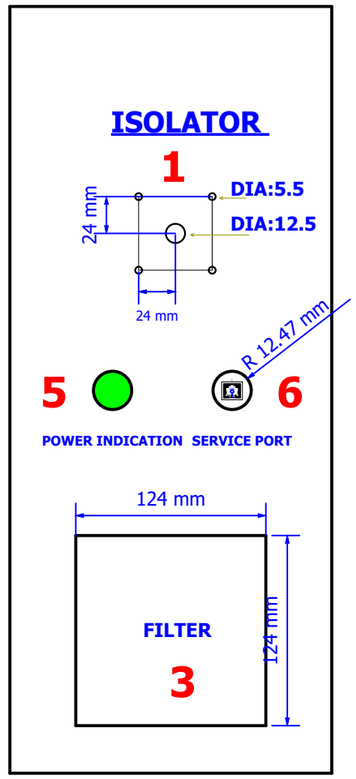
LEFT SIDE VIEW



FRONT SIDE VIEW



RIGHT SIDE VIEW

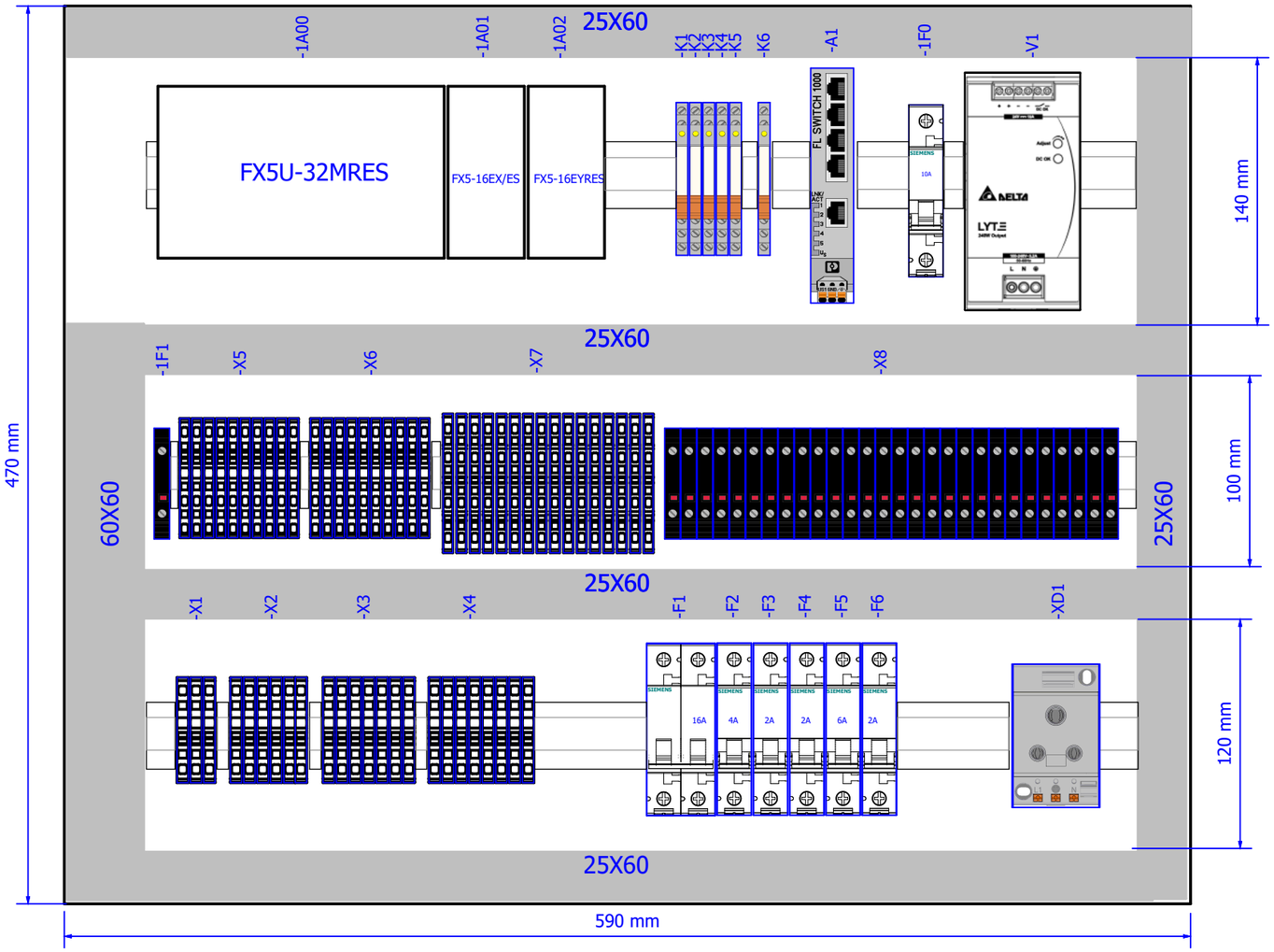


1	ISOLATOR CUTOUT
2	PANEL FAN CUTOUT
3	PANEL FILTER CUTOUT
4	FIELD INTERFACE CUTOUT
5	POWER INDICATION
6	SERVICE PORT

REVISION			Date	24-07-2023	ROBIS	Motherson Automotive & Technology engineering	PANEL CUTOUT		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 3
Modification	Date	Name	Original		Replacement of	Replaced by			Page 4 / 31

LAYOUT DIAGRAM

-1A00	CAAAZ00000000077 PLC FX5U32MRES MITSUBHISHI	1
-1A01	CAAAZ00000000072 16POINT INPUT CARD FX5-16EX/ES -MITSUBHISHI	1
-1A02	CAAAZ00000000073 16POINTS OUTCARD :-FX5-16EYRES, MISTUBHISHI	1
-K1 TO -K6	BMELES0935 SLIM RELAY COIL 24VDC	6
-A1	BMELES1310 ETEHRNET SWITCH 100 MB/S	1
-1F0	BMELES1708 MCB SINGLE POLE 10A SIEMENS MAKE	1
-V1	BMELES1532 DELTA LITE POWER SUPPLY 10A ,1 Phase ,240W	1
-1F1	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	1
-X1	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	2
-X1	BMELES0595 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT YE/GN	1
-X2	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	6
-X3	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	7
-X4	BMELES0595 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT YE/GN	8
-X5	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	10
-X6	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	10
-X7	BMELES0366 TERMINAL BLOCK 2.5 SQ MM 2 IN 2 OUT	16
-X8	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	29
-F1	BMELES3072 SIEMENS MCB 2POLE 16AMPS 5SL62167RC	1
-F2	BMELES1706 MCB SINGLE POLE 4A SIEMENS MAKE	1
-F3	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-F4	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-F5	BMELES1707 MCB SINGLE POLE 6A SIEMENS MAKE	1
-F6	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
XD1	CAABA00000000433 804000 INDIAN 6A DIN RAIL POWER SOCKET	1



REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	LAYOUT DIAGRAM	=
			Ed.	MANORAJ.SELVARAJ				+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856 Page 4
Modification	Date	Name	Original		Replacement of	Replaced by		Page 5 / 31

PROJECT ARCHITECTURE



CAAAZ00000000077
PLC FX5U32MRES
MITSUBHISHI



CAAAZ00000000072
16POINT INPUT CARD
MODEL :-FX5-16EX/ES
MITSUBHISHI



CAAAZ00000000073
16POINTS OUTCARD
MODEL:-FX5-16EYRES,
MISTUBHISHI



BMELES1540
Mushroom Red Turn
Teknic 1NC 2PSML4 01
(2AML4+S2)



TOWERLIGHT 2
STG RG24V BUZ MNT
BMELES0523



BMELES1552
MUSHROOM PB GREEN
Teknic 1NO 2PSM3 10
(2AM3+S1)



BMELES1552
MUSHROOM PB GREEN
Teknic 1NO 2PSM3 10
(2AM3+S1)



BMELES1542
NONIII. YELLOW PUSH
BUTTON, Teknic 1 NO
2PSF8 10 (2AF8+S1)



BMELES1310
Ethernet Switch
100 MB/S



CAAAI00000000586
HMI MODEL : GS2107-WTBD
MITSUBHISH



BMELES3959
Data logic power
scanner PD9531



SENSORS
(CUSTOMER
SCOPE)



BMPNES5128
NEBU-M8G3-K-5-LE3

SCREW DRIVER 1



(CUSTOMER
SCOPE)

INPUT/OUTPUT

SAFETY

ETHERNET COMMUNICATION

RS 232 COMMUNICATION

REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	PROJECT ARCHITECTURE		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV LIGHT STATION-3				FMSPM00856	Page 5
Modification	Date	Name	Original		Replacement of	Replaced by				Page 6 / 31

MAIN POWER DISTRIBUTION



-F1
BMELES3072
SIEMENS MCB 2POLE 16AMPS
5SL62167RC



-Q1
CAABA00000000274
16 A /3P /1WAY ISOLATOR
ROTARY SWITCH

-X1
BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-W0
3Gx2.5 mm²
BMELES1940

BMELES3408
INDUSTRIAL
SOCKET LEGRAND

INDUSTRIAL SOCKET

POWER SUPPLY
1 PHASE 230V
50-60 Hz

-X2
BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-X3
BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-X4
BMELES0595
TERMINAL BLOCK
4.0 SQ MM 1 IN 1 OUT YE/GN

HMI PANEL EARTH

PNEUMATIC PANEL EARTH

PANEL EARTH

5 7

REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	MAIN POWER DISTRIBUTION	=	
			Ed.	MANORAJ.SELVARAJ				+ CP	
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 6
Modification	Date	Name	Original		Replacement of	Replaced by			Page 7 / 31

POWER SUPPLY

1L1-1 / 6.2 -X2:1:B
1N1-1 / 6.5 -X3:1:B
PE-1 / 6.7 -X4:1:B
2.5 mm²
BK
1L1-1
-F2
MCB SINGLE POLE
4 A SIEMENS MAKE
5SL6104RC;
BMELES1706
2L1-1
1N1-1
PE-1
-V1
BMELES1532
DELTA LITE POWER SUPPLY
10A ,1 Phase ,240W
Weidmüller
PROeco
L(+) N(-) PE
Adjust
22-28 V
+ + - -
13 14
1.5 mm²
2.5 mm²
DBU
GNYE
24VDC
P0VDC
PE-3
-1F0:2 10.2 / 24VDC
-X6:1 10.6 / P0VDC
-X4:3:B 6.8 / PE-3

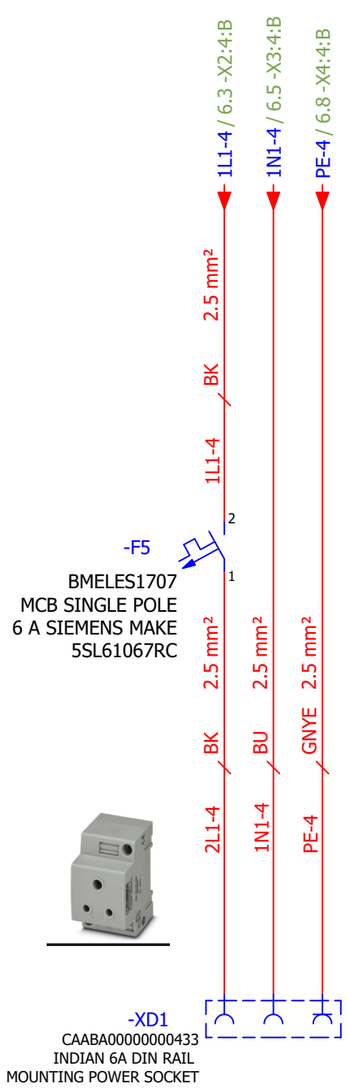
1L1-2 / 6.3 -X2:2:B
1N1-2 / 6.5 -X3:2:B
PE-2 / 6.7 -X4:2:B
1L1-2
-F3
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC
BK/1
0.75 mm²
3L1-2
-W1
BMELES0075
3 G 0.75 mm²
BK/2
GNYE
2L1-2
1N1-2
PE-2
-M1
M
1~
BMELES0049
SK 3322107-
FAN AND FILTER UNIT
-LS
BMELES1669
AE PANEL DOOR
LIMIT SWITCH -
SZ 4127010
21
22
4L1-2
-W2
BMELES0075
3 G 0.75 mm²
-H1
BMELES2527
1Feet 6W LED
TUBE LIGHT WITH
ALUMINIUM BATTERY (CW)

1N1-3 / 6.5 -X3:3:B
1L1-3 / 6.3 -X2:3:B
1L1-3
-F4
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC
1 mm²
1.5 mm²
BU
1N1-3
2L1-3
-K6:11 21.1 / 2L1-3

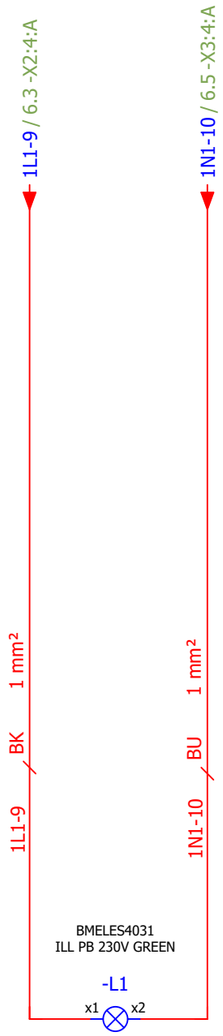
6 8

REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	POWER SUPPLY		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 7
Modification	Date	Name	Original		Replacement of	Replaced by			Page 8 / 31

INDICATION LAMP



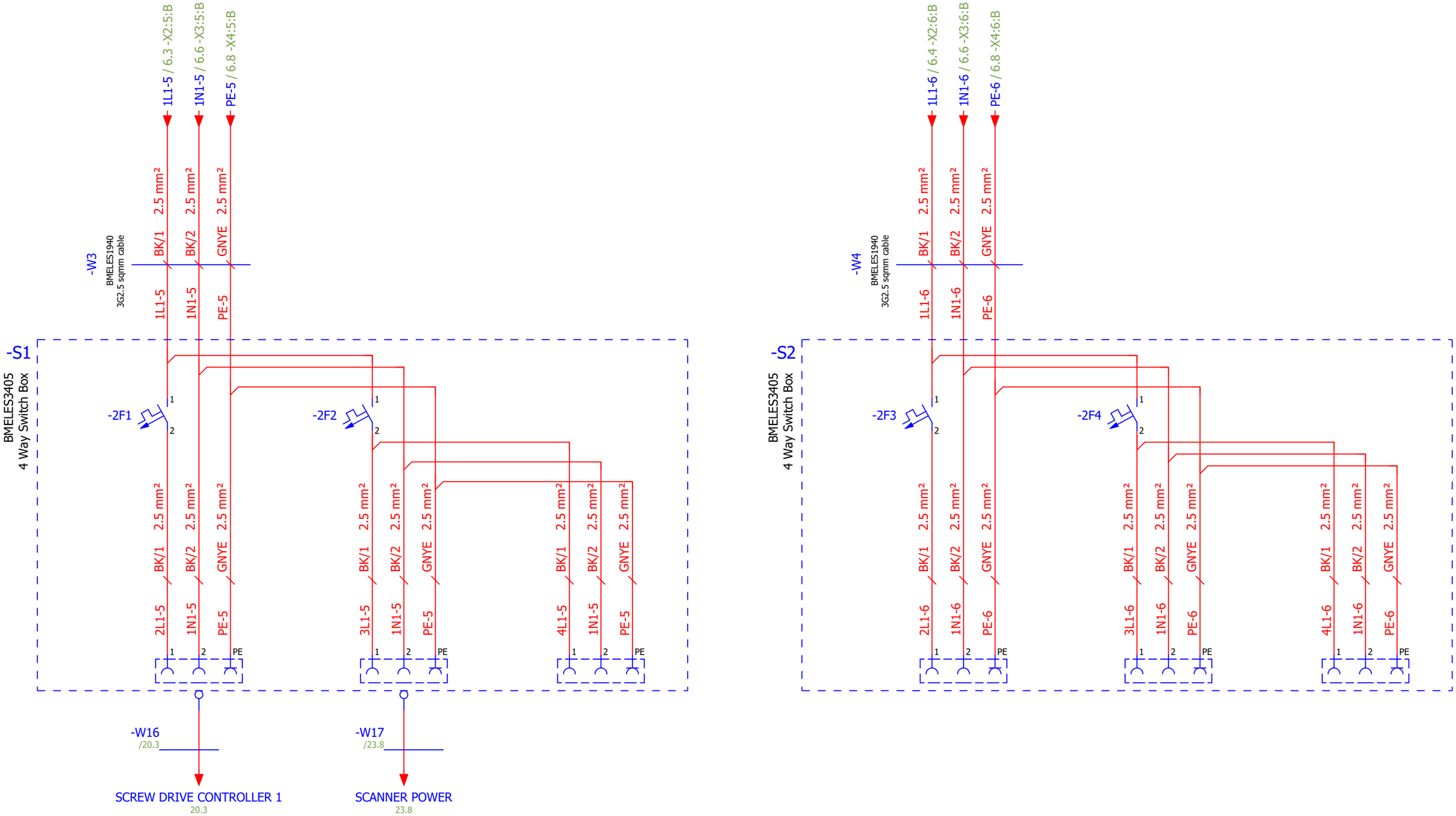
POWER SOCKET



230V GREEN
POWER INDICATION
LAMP

REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	INDICATION LAMP		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 8
Modification	Date	Name	Original		Replacement of	Replaced by			Page 9 / 31

SOCKET INTERFACE

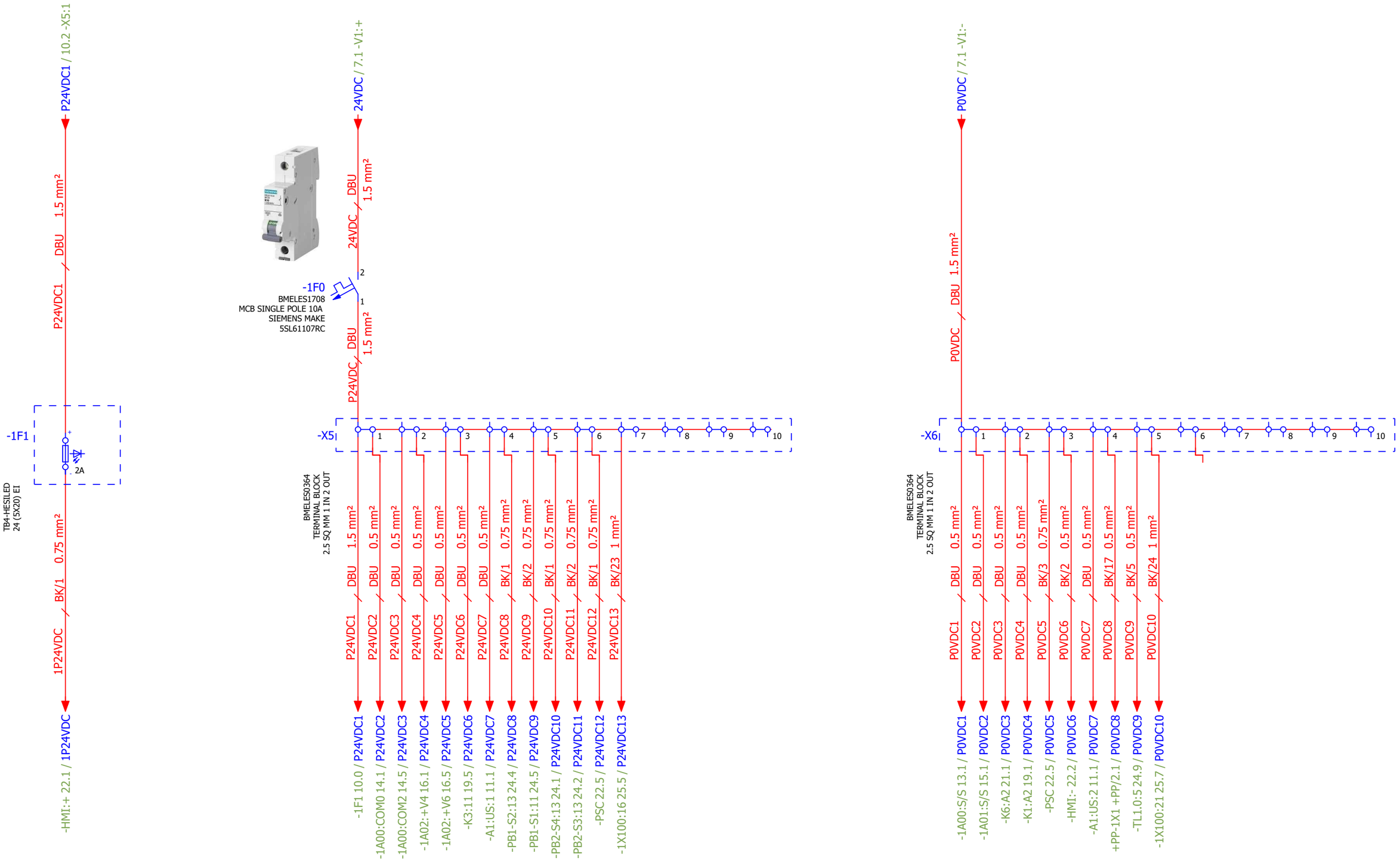


8 10

REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	SOCKET INTERFACE		=	
			Ed.	MANORAJ.SELVARAJ					+ CP	
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 9	
Modification	Date	Name	Original		Replacement of	Replaced by			Page 10 / 31	

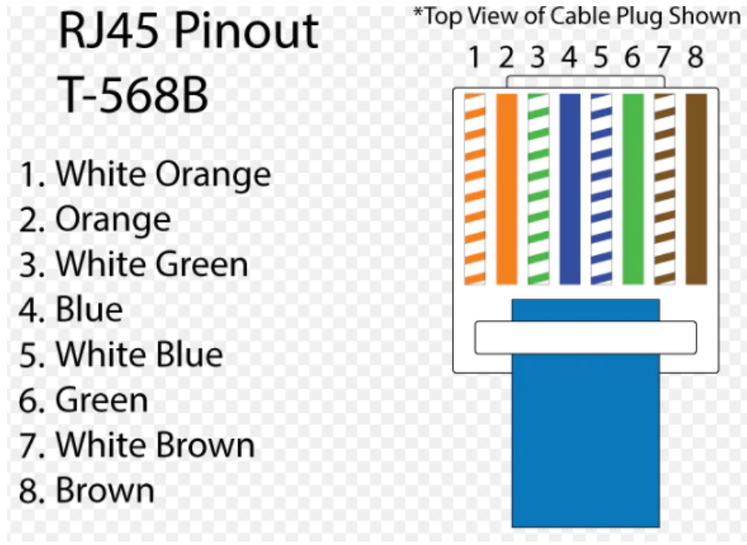
0	1	2	3	4	5	6	7	8	9

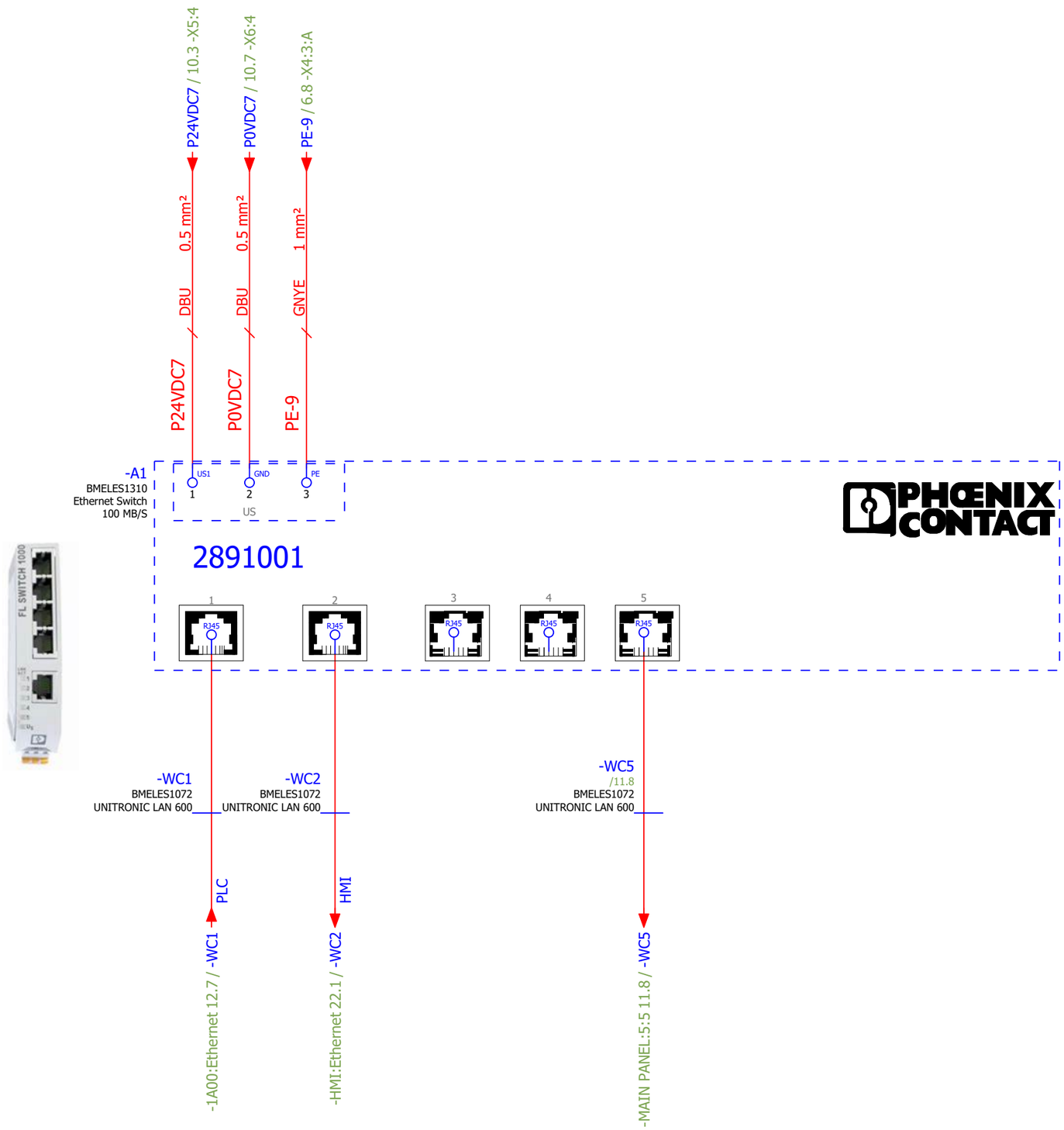
DISTRIBUTION 24VDC TERMINAL BLOCK

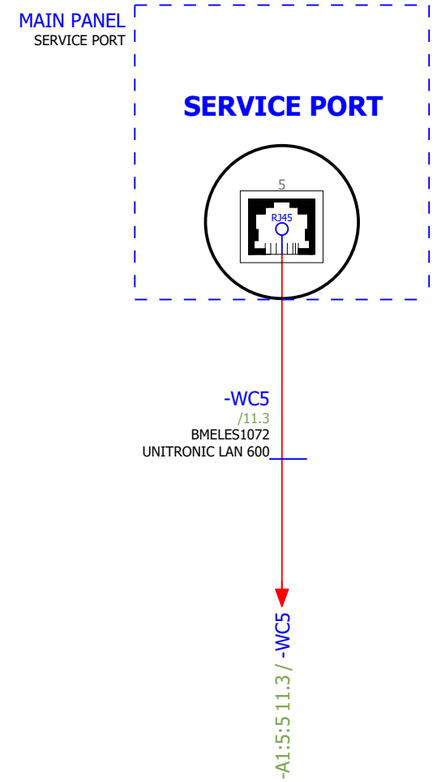


REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	DISTRIBUTION 24VDC TERMINAL BLOCK	=
			Ed.	MANORAJ.SELVARAJ				+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856 Page 10
Modification	Date	Name	Original		Replacement of	Replaced by		Page 11 / 31

COMMUNICATION







REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	COMMUNICATION		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 11
Modification	Date	Name	Original		Replacement of	Replaced by			Page 12 / 31

MAIN PLC POWER SUPPLY



-F6
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC

1L1-7 / 6.3 -X2:2:A
1N1-7 / 6.6 -X3:7:B
PE-7 / 6.9 -X4:7:B

BK 2.5 mm²
1L1-7
2
1
2.5 mm²
BK
2L1-7
2.5 mm²
BU
1N1-7
2.5 mm²
GNYE
PE-7

WC1 / 11.1 -A1:1:1

-WC1
/11.1
/22.1
BMELES1072
2170144-UNITRONIC
LAN 600S/FTP CAT7Y



-1A00
CAAAZ00000000077
PLC
FX5U32MRES
MAKE :- MITSUBHISHI

L | N | PE | SG (GND) | SDB (TXD-) | SDA (TXD+) | RDB (RXD-) | RDA (RXD+) | V1+ (PEW0) | V2+ (PEW2) | V- (PEW4) | V+ (PAW0) | V- (PAW2) | Ethernet

Ethernet port

MITSUBISHI ELECTRIC
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 1 of 5

REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	MAIN PLC POWER SUPPLY		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 12
Modification	Date	Name	Original		Replacement of	Replaced by			Page 13 / 31

MAIN INPUT MODULE



P0VDC1 / 10.6 / -X6:1

P0VDC1 DBU 0.5 mm²

-1A00
CAAAZ00000000077
PLC FX5U32MRES
MAKE :- MITSUBHISHI

S/S	0V	24V	X00	X01	X02	X03	X04	X05	X06	X07
			E0.0	E0.1	E0.2	E0.3	E0.4	E0.5	E0.6	E0.7
			EMERGENCY STOP PB	RH CYCLE START PB			AIR PRESSURE SWITCH		SCREW DRIVER OK ALL	
				LH CYCLE START PB		RESET		SPARE		SCREW DRIVER NG

DBU 0.5 mm²

X00 / 17.0 -X7:1:A
X01 / 17.0 -X7:1:C
X02 / 17.0 -X7:2:A
X03 / 17.1 -X7:2:C
X04 / 17.1 -X7:3:A
X05 / 17.1 -X7:3:C
X06 / 17.1 -X7:4:A
X07 / 17.1 -X7:4:C

MITSUBISHI ELECTRIC
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 2 of 5

X10 / 17.1 -X7:5:A
X11 / 17.1 -X7:5:C
X12 / 17.2 -X7:6:A
X13 / 17.2 -X7:6:C
X14 / 17.2 -X7:7:A
X15 / 17.2 -X7:7:C
X16 / 17.2 -X7:8:A
X17 / 17.2 -X7:8:C

-1A00

X10	X11	X12	X13	X14	X15	X16	X17			
E1.0	E1.1	E1.2	E1.3	E1.4	E1.5	E1.6	E1.7			
SPARE		CLAMPING CYL1 FWD RS		CLAMPING CYL2 FWD RS		FIXTURE SPARE				
	SPARE		CLAMPING CYL1 REV RS		CLAMPING CYL2 REV RS		FIXTURE SPARE			

MITSUBISHI ELECTRIC
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 3 of 5

REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	MAIN INPUT MODULE	=
			Ed.	MANORAJ.SELVARAJ				+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856 Page 13
Modification	Date	Name	Original		Replacement of	Replaced by		Page 14 / 31

MAIN OUTPUT MODULE



-1A00
CAAAZ00000000077
PLC FX5U32MRES
MAKE :- MITSUBHISHI

Mitsubishi FX5U-32MR-ES 16DI/16DO Part 4 of 5

MITSUBISHI ELECTRIC

	TOWER LAMP YELLOW	TOWER LAMP BUZZER			MACHINE LIGHT			SCREW DRIVER RESET
TOWER LAMP GREEN		TOWER LAMP RED			SHUTOFF VALVE			SCREW DRIVER ENABLE
COM0	A0.0 Y00	A0.1 Y01	A0.2 Y02	A0.3 Y03	COM1	A0.4 Y04	A0.5 Y05	A0.6 Y06	A0.7 Y07

0.5 mm² DBU P24VDC2 → -X5:1 10.2 / P24VDC2

0.5 mm² DBU Y00	0.5 mm² DBU Y01	0.5 mm² DBU Y02	0.5 mm² DBU Y03	0.5 mm² DBU P24VDC2	0.5 mm² DBU Y04	0.5 mm² DBU Y05	0.5 mm² DBU Y06	0.5 mm² DBU Y07
-X8:1:A 18.1 / Y00	-X8:2:A 18.1 / Y01	-X8:3:A 18.1 / Y02	-X8:4:A 18.1 / Y03		-X8:5:A 18.2 / Y04	-K6:A1 21.0 / Y05	-K1:A1 19.0 / Y06	-K2:A1 19.2 / Y07

-1A00
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 5 of 5

MITSUBISHI ELECTRIC

SPARE		CLAMPING CYL1 REV				CLAMPING CYL2 REV		SPARE VALVE
	SPARE		CLAMPING CYL1 FWD				CLAMPING CYL2 FWD	SPARE VALVE
COM2	A1.0 Y10	A1.1 Y11	A1.2 Y12	A1.3 Y13	COM3	A1.4 Y14	A1.5 Y15	A1.6 Y16	A1.7 Y17

0.5 mm² DBU P24VDC3 → -X5:2 10.2 / P24VDC3

0.5 mm² DBU Y10	0.5 mm² DBU Y11	0.5 mm² DBU Y12	0.5 mm² DBU Y13	0.5 mm² DBU P24VDC3	0.5 mm² DBU Y14	0.5 mm² DBU Y15	0.5 mm² DBU Y16	0.5 mm² DBU Y17
-X8:6:A 18.2 / Y10	-X8:7:A 18.2 / Y11	-X8:8:A 18.2 / Y12	-X8:9:A 18.2 / Y13		-X8:10:A 18.3 / Y14	-X8:11:A 18.3 / Y15	-X8:12:A 18.3 / Y16	-X8:13:A 18.3 / Y17

REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	MAIN OUTPUT MODULE	=
			Ed.	MANORAJ.SELVARAJ				+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3		FMSPM00856	Page 14
Modification	Date	Name	Original		Replacement of	Replaced by		Page 15 / 31

EXPANSION INPUT MODULE



P0VDC2 / 10.6 -X6:1 — DBU 0.5 mm² — P0VDC2 — S/S

X20 / 17.3 -X7:9:A — DBU 0.5 mm² — X20 — E2.0 — FIXTURE SPARE

X21 / 17.3 -X7:9:C — DBU 0.5 mm² — X21 — E2.1 — FIXTURE SPARE

X22 / 17.3 -X7:10:A — DBU 0.5 mm² — X22 — E2.2 — FIXTURE SPARE

X23 / 17.3 -X7:10:C — DBU 0.5 mm² — X23 — E2.3 — FIXTURE SPARE

X24 / 17.3 -X7:11:A — DBU 0.5 mm² — X24 — E2.4 — FIXTURE SPARE

X25 / 17.3 -X7:11:C — DBU 0.5 mm² — X25 — E2.5 — FIXTURE SPARE

X26 / 17.3 -X7:12:A — DBU 0.5 mm² — X26 — E2.6 — HIGH VARIANT

X27 / 17.4 -X7:12:C — DBU 0.5 mm² — X27 — E2.7 — MID VARIANT

-1A01
/15.5

CAAAZ00000000072
16POINTINPUTCARD
MODEL :-FX5-16EX/ES
MAKE :-MITSUBHISHI

MITSUBISHI ELECTRIC

Mitsubishi | FX5U-EX/ES | 16DI | Part 1 of 2

X30 / 17.4 -X7:13:A — DBU 0.5 mm² — X30 — E3.0 — LOW VARIANT

X31 / 17.4 -X7:13:C — DBU 0.5 mm² — X31 — E3.1 — SPARE

X32 / 17.4 -X7:14:A — DBU 0.5 mm² — X32 — E3.2 — SPARE

X33 / 17.4 -X7:14:C — DBU 0.5 mm² — X33 — E3.3 — SPARE

X34 / 17.4 -X7:15:A — DBU 0.5 mm² — X34 — E3.4 — SPARE

X35 / 17.5 -X7:15:C — DBU 0.5 mm² — X35 — E3.5 — SPARE

X36 / 17.5 -X7:16:A — DBU 0.5 mm² — X36 — E3.6 — SPARE

X37 / 17.5 -X7:16:C — DBU 0.5 mm² — X37 — E3.7 — SPARE

-1A01
/15.1

MITSUBISHI ELECTRIC

Mitsubishi | FX5U-EX/ES | 16DI | Part 2 of 2

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	EXPANSION INPUT MODULE	=
			Ed.	manoraj.selvaraj	S220 HL HV & LV LIGHT STATION-3			+ CP
			Appr.				FMSPM00856	Page 15
Modification	Date	Name	Original		Replacement of	Replaced by		Page 16 / 31

EXPANSION OUTPUT MODULE



-1A02
/16.5
CAAAZ00000000073
16POINTS OUTCARD
-FX5-16EYRES, MAKE
MISTUBHISHI

Mitsubishi FX5-16EYR-ES 16DO Part 1 of 2

MITSUBISHI ELECTRIC

SPARE VALVE SPARE SPARE SPARE

SPARE VALVE SPARE SPARE SPARE

+V4 Y20 Y21 Y22 Y23 Y24 Y25 Y26 Y27

0.5 mm² DBU P24VDC4

0.5 mm² DBU Y20 → -X8:14:A 18.4 / Y20

0.5 mm² DBU Y21 → -X8:15:A 18.4 / Y21

0.5 mm² DBU Y22 → -X8:16:A 18.4 / Y22

0.5 mm² DBU Y23 → -X8:17:A 18.4 / Y23

0.5 mm² DBU Y24 → -X8:18:A 18.4 / Y24

0.5 mm² DBU Y25 → -X8:19:A 18.5 / Y25

0.5 mm² DBU Y26 → -X8:20:A 18.5 / Y26

0.5 mm² DBU Y27 → -X8:21:A 18.5 / Y27

P24VDC4 → -X5:2 10.3 / P24VDC4

-1A02
/16.1

Mitsubishi FX5-16EYR-ES 16DO Part 2 of 2

MITSUBISHI ELECTRIC

SPARE SPARE SPARE SPARE

SPARE SPARE SPARE SPARE

+V6 Y30 Y31 Y32 Y33 Y34 Y35 Y36 Y37

0.5 mm² DBU P24VDC5

0.5 mm² DBU Y30 → -X8:22:A 18.5 / Y30

0.5 mm² DBU Y31 → -X8:23:A 18.5 / Y31

0.5 mm² DBU Y32 → -X8:24:A 18.6 / Y32

0.5 mm² DBU Y33 → -X8:25:A 18.6 / Y33

0.5 mm² DBU Y34 → -X8:26:A 18.6 / Y34

0.5 mm² DBU Y35 → -X8:27:A 18.6 / Y35

0.5 mm² DBU Y36 → -X8:28:A 18.7 / Y36

0.5 mm² DBU Y37 → -X8:29:A 18.7 / Y37

P24VDC5 → -X5:3 10.3 / P24VDC5

15 17

REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	EXPANSION OUTPUT MODULE	=
			Ed.	MANORAJ.SELVARAJ				+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3		FMSPM00856	Page 16
Modification	Date	Name	Original		Replacement of	Replaced by		Page 17 / 31

BMELES2970
TERMINAL BLOCK CXDL2.5
2.5 SQ MM 2 IN 2 OUT
-X7-

-PB1-S1:12 24.5 / X00	X00	BK/4	0.75 mm²	1	X00	DBU	0.5 mm²	X00 / 13.2 -1A00:X00
-PB2-S4:14 24.1 / X01	X01	BK/3	0.75 mm²		X01	DBU	0.5 mm²	X01 / 13.3 -1A00:X01
-PB1-S2:14 24.4 / X02	X02	BK/3	0.75 mm²	2	X02	DBU	0.5 mm²	X02 / 13.3 -1A00:X02
-PB2-S3:14 24.2 / X03	X03	BK/4	0.75 mm²		X03	DBU	0.5 mm²	X03 / 13.3 -1A00:X03
-PSC 22.6 / X04	X04	BK/2	0.75 mm²	3	X04	DBU	0.5 mm²	X04 / 13.4 -1A00:X04
					X05	DBU	0.5 mm²	X05 / 13.4 -1A00:X05
-K4:14 19.7 / X06	X06	DBU	0.5 mm²	4	X06	DBU	0.5 mm²	X06 / 13.4 -1A00:X06
-K5:14 19.9 / X07	X07	DBU	0.5 mm²		X07	DBU	0.5 mm²	X07 / 13.5 -1A00:X07
				5	X10	DBU	0.5 mm²	X10 / 13.6 -1A00:X10
					X11	DBU	0.5 mm²	X11 / 13.6 -1A00:X11
-1X100:1 25.1 / X12	X12	BK/1	1 mm²	6	X12	DBU	0.5 mm²	X12 / 13.7 -1A00:X12
-1X100:2 25.1 / X13	X13	BK/2	1 mm²		X13	DBU	0.5 mm²	X13 / 13.7 -1A00:X13
-1X100:3 25.1 / X14	X14	BK/3	1 mm²	7	X14	DBU	0.5 mm²	X14 / 13.8 -1A00:X14
-1X100:4 25.2 / X15	X15	BK/4	1 mm²		X15	DBU	0.5 mm²	X15 / 13.8 -1A00:X15
-1X100:5 25.2 / X16	X16	BK/5	1 mm²	8	X16	DBU	0.5 mm²	X16 / 13.8 -1A00:X16
-1X100:6 25.2 / X17	X17	BK/6	1 mm²		X17	DBU	0.5 mm²	X17 / 13.9 -1A00:X17
-1X100:7 25.3 / X20	X20	BK/7	1 mm²	9	X20	DBU	0.5 mm²	X20 / 15.1 -1A01:X20
-1X100:8 25.3 / X21	X21	BK/8	1 mm²		X21	DBU	0.5 mm²	X21 / 15.2 -1A01:X21
-1X100:9 25.3 / X22	X22	BK/9	1 mm²	10	X22	DBU	0.5 mm²	X22 / 15.2 -1A01:X22
-1X100:10 25.3 / X23	X23	BK/10	1 mm²		X23	DBU	0.5 mm²	X23 / 15.3 -1A01:X23
-1X100:11 25.4 / X24	X24	BK/11	1 mm²	11	X24	DBU	0.5 mm²	X24 / 15.3 -1A01:X24
-1X100:12 25.4 / X25	X25	BK/12	1 mm²		X25	DBU	0.5 mm²	X25 / 15.3 -1A01:X25
-1X100:13 25.4 / X26	X26	BK/13	1 mm²	12	X26	DBU	0.5 mm²	X26 / 15.4 -1A01:X26
-1X100:14 25.5 / X27	X27	BK/14	1 mm²		X27	DBU	0.5 mm²	X27 / 15.4 -1A01:X27
-1X100:15 25.5 / X30	X30	BK/15	1 mm²	13	X30	DBU	0.5 mm²	X30 / 15.6 -1A01:X30
					X31	DBU	0.5 mm²	X31 / 15.6 -1A01:X31
				14	X32	DBU	0.5 mm²	X32 / 15.6 -1A01:X32
					X33	DBU	0.5 mm²	X33 / 15.7 -1A01:X33
				15	X34	DBU	0.5 mm²	X34 / 15.7 -1A01:X34
					X35	DBU	0.5 mm²	X35 / 15.7 -1A01:X35
				16	X36	DBU	0.5 mm²	X36 / 15.8 -1A01:X36
					X37	DBU	0.5 mm²	X37 / 15.8 -1A01:X37

INPUT TERMINAL DISTRIBUTION

Modification	REVISION		
Date			
Name			
Original	Appr.	Ed.	Date
	manoraj.selvaraj		01-08-2023
Replacement of	ROBIS		
Replaced by	S220 HL HV & LV LIGHT STATION-3		
	Motherson Automotive & Technology engineering		
	INPUT TERMINAL DISTRIBUTION		
	FMSPM00856		Page 17
		+ CP	Page 18 / 31

OUTPUT TERMINAL DISTRIBUTION

ALL FUSE SHOULD BE 0.5 A



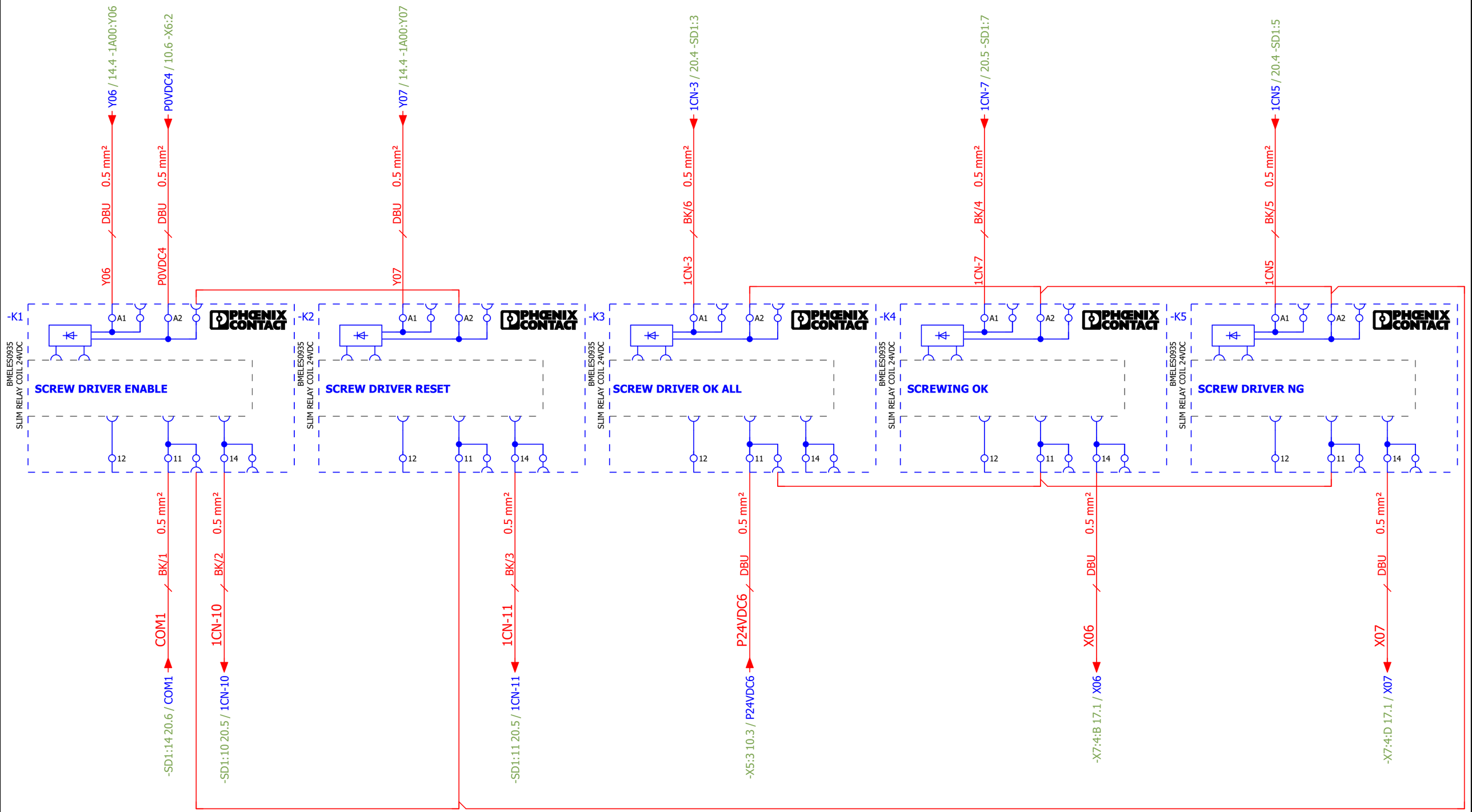
-X8
BMELES2968
TERMINAL BLOCK 2.5 SQ MM
1 IN 2 OUT

Y00 / 14.1 / -1A00:Y00	Y01 / 14.2 / -1A00:Y01	Y02 / 14.2 / -1A00:Y02	Y03 / 14.3 / -1A00:Y03	Y04 / 14.3 / -1A00:Y04	Y10 / 14.6 / -1A00:Y10	Y11 / 14.6 / -1A00:Y11	Y12 / 14.7 / -1A00:Y12	Y13 / 14.7 / -1A00:Y13	Y14 / 14.8 / -1A00:Y14	Y15 / 14.8 / -1A00:Y15	Y16 / 14.8 / -1A00:Y16	Y17 / 14.9 / -1A00:Y17	Y20 / 16.1 / -1A02:Y20	Y21 / 16.2 / -1A02:Y21	Y22 / 16.2 / -1A02:Y22	Y23 / 16.3 / -1A02:Y23	Y24 / 16.3 / -1A02:Y24	Y25 / 16.3 / -1A02:Y25	Y26 / 16.4 / -1A02:Y26	Y27 / 16.4 / -1A02:Y27	Y30 / 16.6 / -1A02:Y30	Y31 / 16.6 / -1A02:Y31	Y32 / 16.6 / -1A02:Y32	Y33 / 16.7 / -1A02:Y33	Y34 / 16.7 / -1A02:Y34	Y35 / 16.8 / -1A02:Y35	Y36 / 16.8 / -1A02:Y36	Y37 / 16.8 / -1A02:Y37
0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²
DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU
Y00	Y01	Y02	Y03	Y04	Y10	Y11	Y12	Y13	Y14	Y15	Y16	Y17	Y20	Y21	Y22	Y23	Y24	Y25	Y26	Y27	Y30	Y31	Y32	Y33	Y34	Y35	Y36	Y37		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29		
0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A		
OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE		

Y00	Y01	Y02	Y03	Y04	Y12	Y13	Y14	Y15	Y16	Y17	Y20	Y21																		
0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²																		
BK/1	BK/2	BK/3	BK/4	BK/1	BK/2	BK/3	BK/4	BK/5	BK/6	BK/7	BK/8	BK/9																		
Y00	Y01	Y02	Y03	Y04	Y12	Y13	Y14	Y15	Y16	Y17	Y20	Y21																		
-TL1.0:1 24.8 / Y00	-TL1.0:2 24.8 / Y01	-TL1.0:3 24.8 / Y02	-TL1.0:4 24.9 / Y03	+PP-1X1:1 +PP/2.1 / Y04	+PP-1X1:2 +PP/2.1 / Y12	+PP-1X1:3 +PP/2.2 / Y13	+PP-1X1:4 +PP/2.2 / Y14	+PP-1X1:5 +PP/2.2 / Y15	+PP-1X1:6 +PP/2.3 / Y16	+PP-1X1:7 +PP/2.3 / Y17	+PP-1X1:8 +PP/2.3 / Y20	+PP-1X1:9 +PP/2.4 / Y21																		

REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	OUTPUT TERMINAL DISTRIBUTION	=
			Ed.	MANORAJ.SELVARAJ				+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856 Page 18
Modification	Date	Name	Original		Replacement of	Replaced by		Page 19 / 31

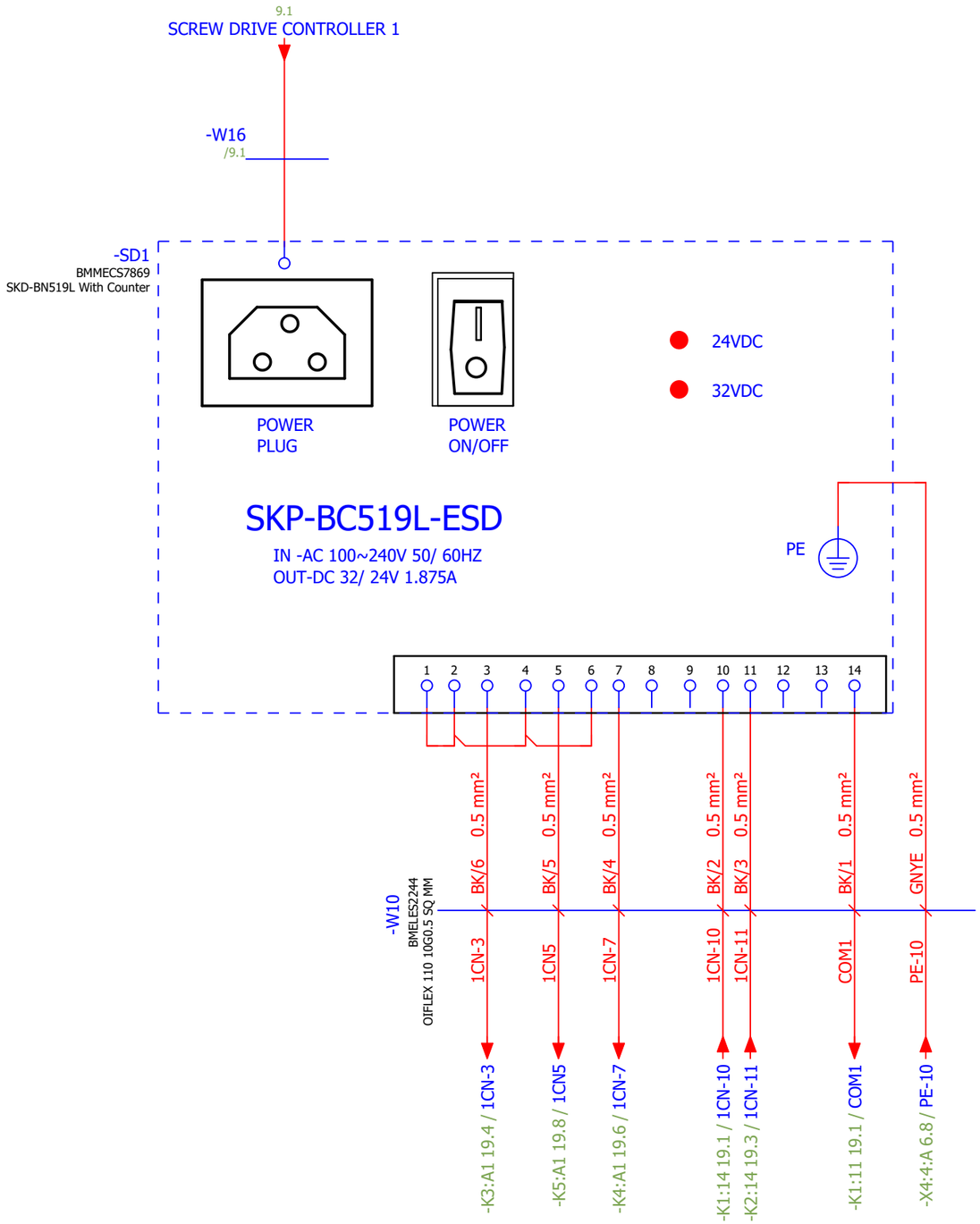
RELAY INTERFACE FOR SCREWING DRIVER 1



18 20

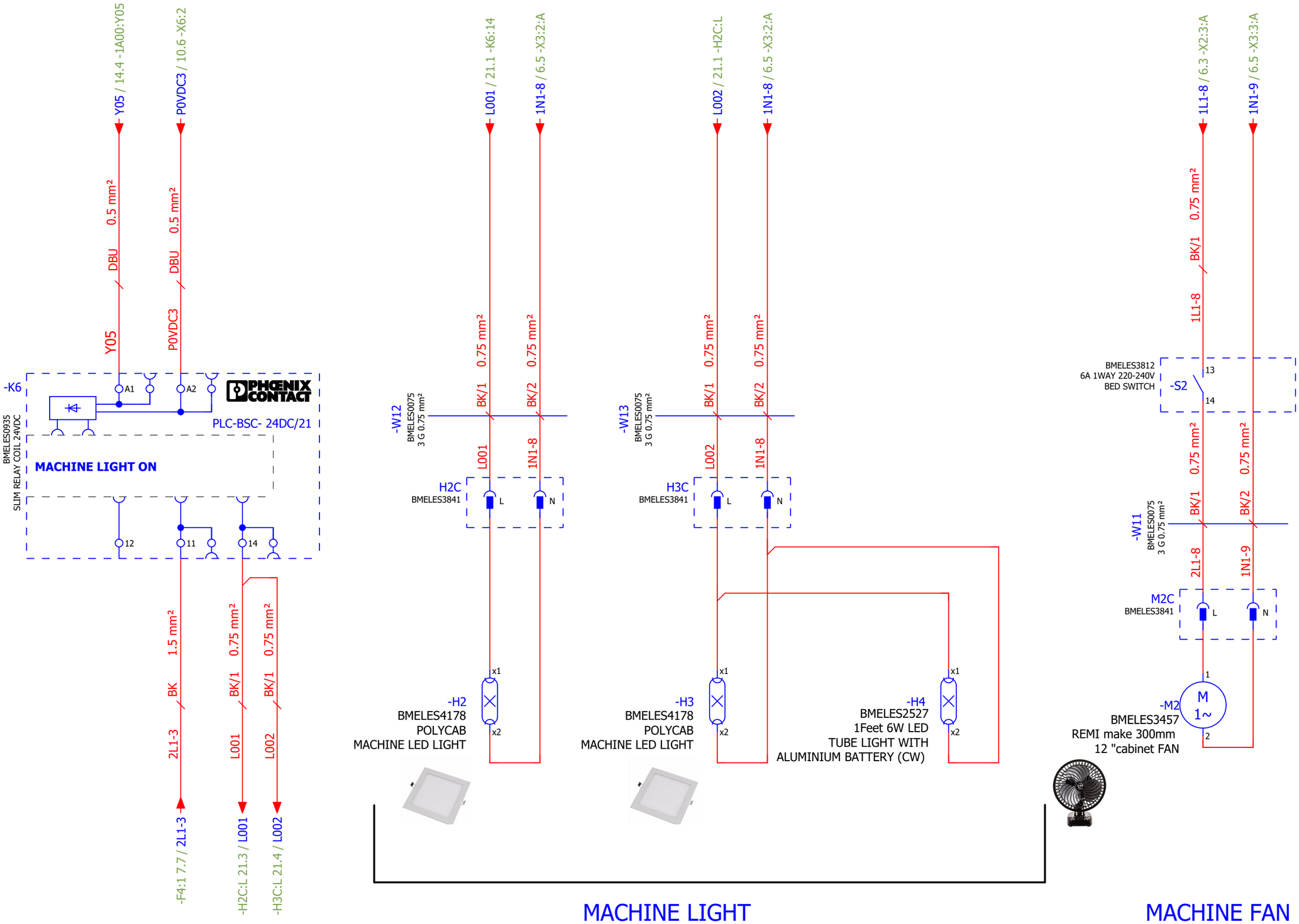
REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	RELAY INTERFACE FOR SCREWING DRIVER 1		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 19
Modification	Date	Name	Original		Replacement of	Replaced by			Page 20 / 31

SCREWING DRIVER INTERFACE 1



REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	SCREWING DRIVER INTERFACE 1		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 20
Modification	Date	Name	Original		Replacement of	Replaced by			Page 21 / 31

MACHINE LIGHT & FAN INTERFACE



REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	MACHINE LIGHT & FAN INTERFACE		=
			Ed.	MANORAJ.SELVARAJ	S220 HL HV & LV LIGHT STATION-3				+ CP
			Appr.					FMSPM00856	Page 21
Modification	Date	Name	Original		Replacement of	Replaced by			Page 22 / 31

HMI INTERFACE



-HMI
CAAAI00000000586
HMI GS2107-WTBD

Ethernet RS-232 RS-422 USB

MITSUBISHI ELECTRIC

+ - FG

-WC1
BMELES1072
UNITRONIC LAN 600

-WC6
BMELES0221
0034504 - Unitronic
LIYCY 4 X 0.34

-WC2 / 11.2 -A1:2:2

-WC6 / 23.2 -HMI RS232-COM1

-W5
BMELES1937
3 G 0.75 mm²

0.75 mm²	0.75 mm²	0.75 mm²
BK/1	BK/2	GNYE
1P24VDC	P0VDC6	PE-8

-1F1 / 10.0 / 1P24VDC

-X6:3 / 10.7 / P0VDC6

-X4:2:A / 6.7 / PE-8

HMI POWER SUPPLY

-W6
BMELES1937
3 G 0.75 mm²

0.75 mm²	0.75 mm²	0.75 mm²
BK/1	BK/3	BK/2
P24VDC12	P0VDC5	X04

P24VDC12 / 10.4 -X5:6

P0VDC5 / 10.7 -X6:3

X04 / 17.1 -X7:3:B

-PSC
3 PIN CONNECTOR

BN BU BK WH

PS1

1 2 3 4

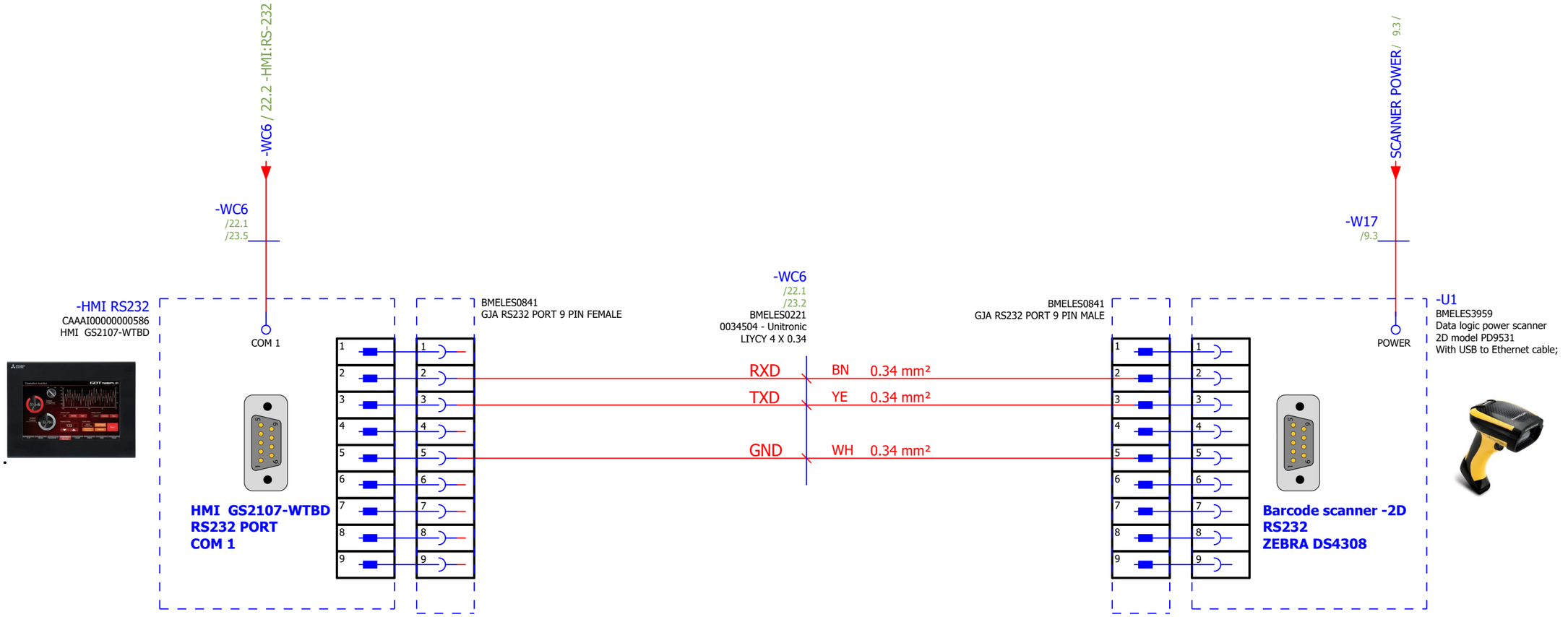
P

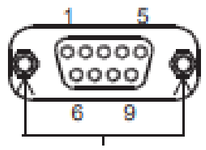


PRESSURE SWITCH

REVISION			Date	24-07-2023	ROBIS	Motherson Automotive & Technology engineering	HMI INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 22
Modification	Date	Name	Original		Replacement of	Replaced by			Page 23 / 31

HMI TO SCANNER COMMUNICATION



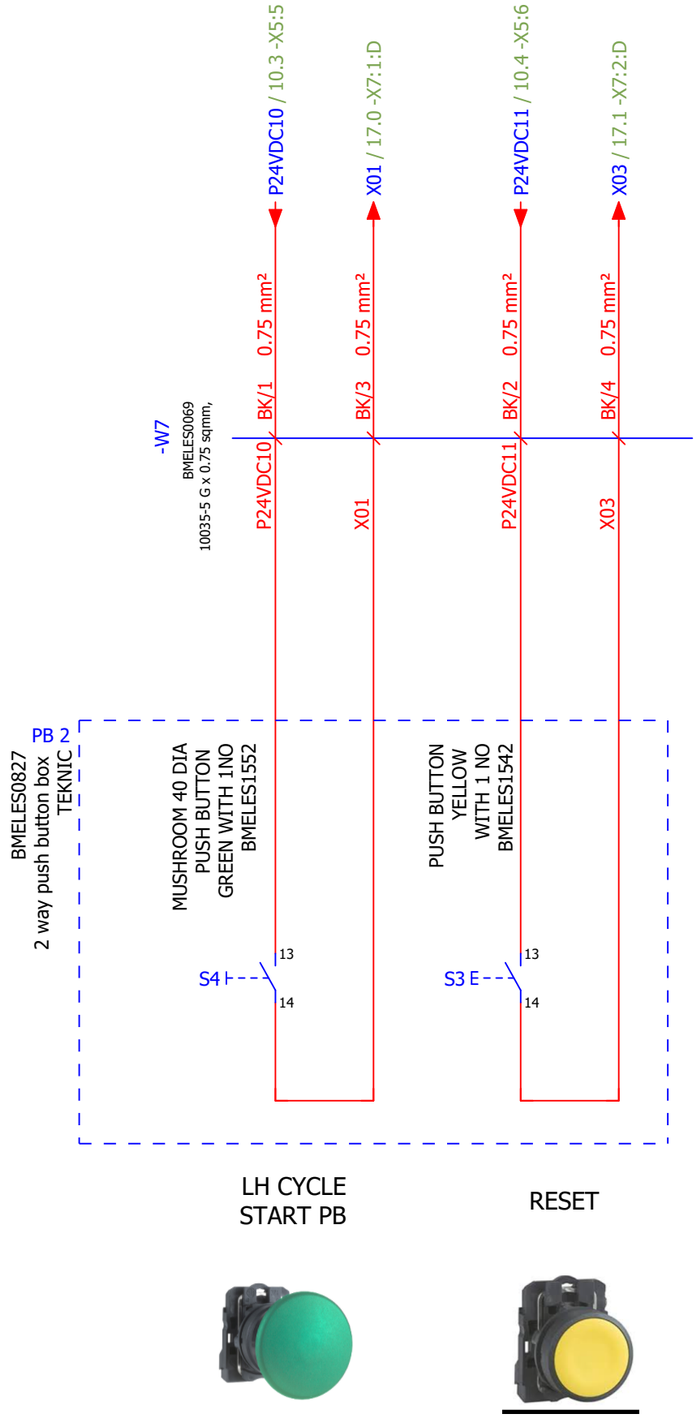


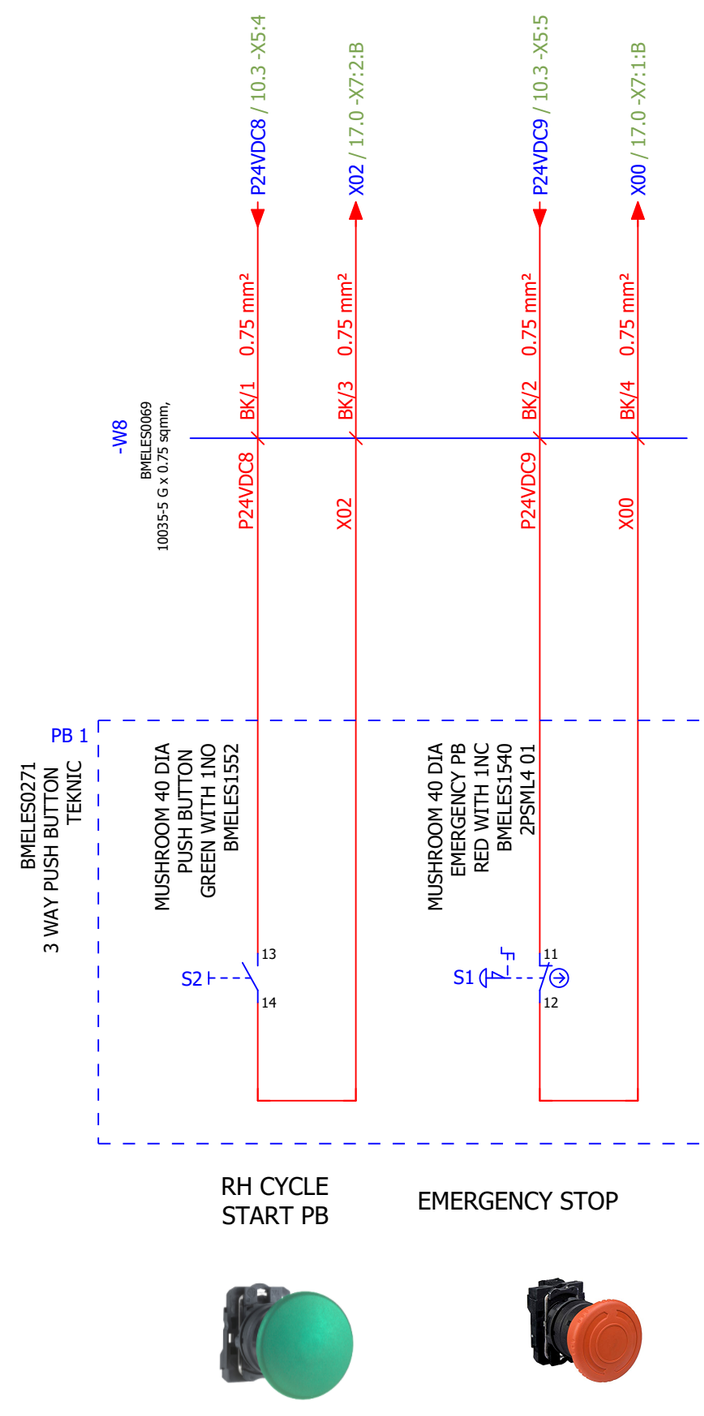
Screws to fix
RS-232C connector
#4-40UNC
(inch screw thread)

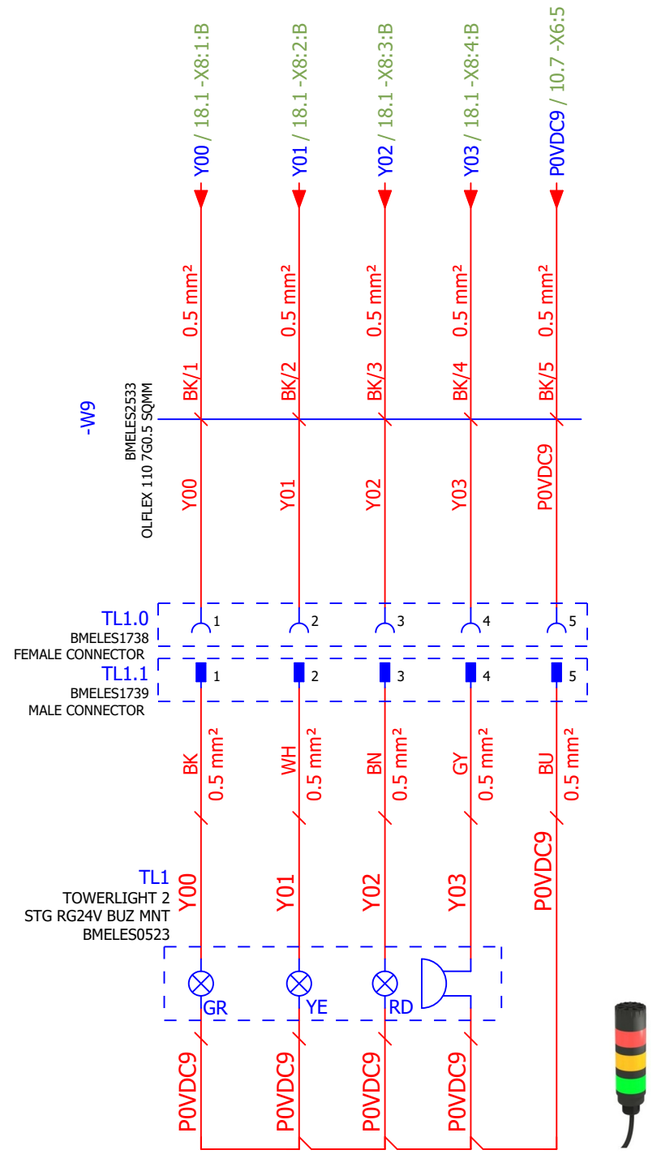
Pin No.	Signal	Name
1	CD (DCD)	Receive carrier detection
2	RD (RXD)	Receive data
3	SD (TXD)	Send data
4	ER (DTR)	Send request
5	SG (GND)	Signal ground
6	DR (DSR)	Send enabled
7, 8, 9		Not used

REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	HMI TO SCANNER COMMUNICATION		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 23
Modification	Date	Name	Original		Replacement of	Replaced by			Page 24 / 31

PUSHBUTTON & TOWER LAMP INTERFACE

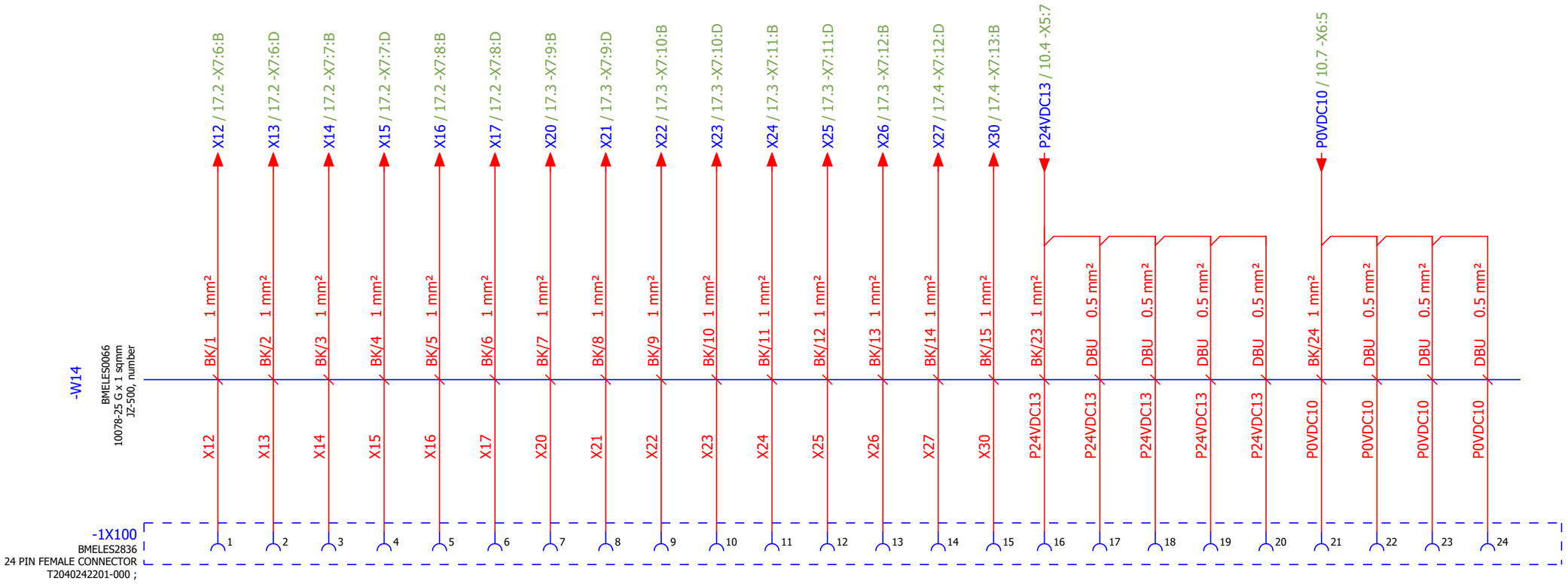






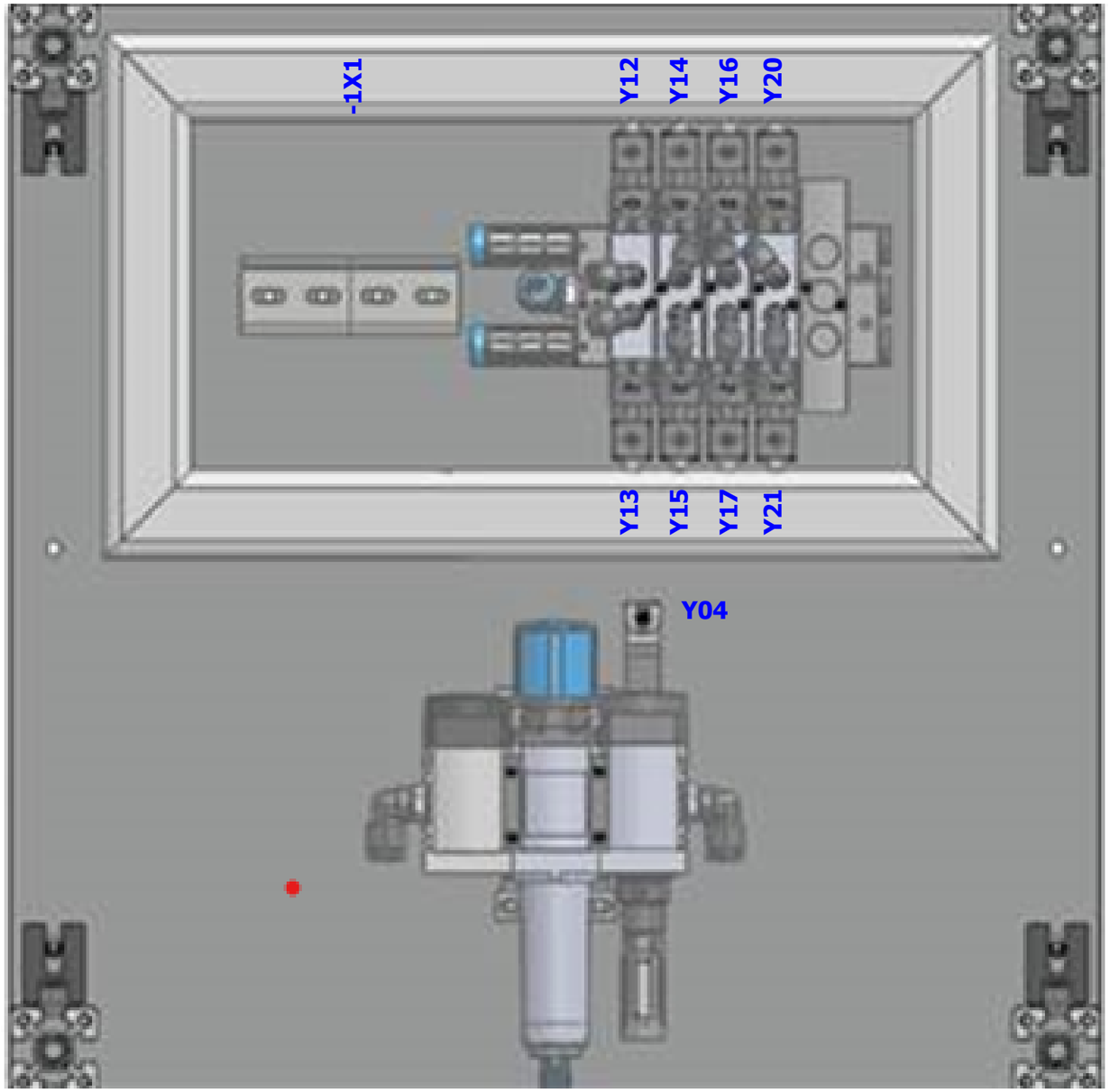
REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	PUSHBUTTON & TOWER LAMP INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 24
Modification	Date	Name	Original		Replacement of	Replaced by			Page 25 / 31

24 PIN I/O INTERFACE



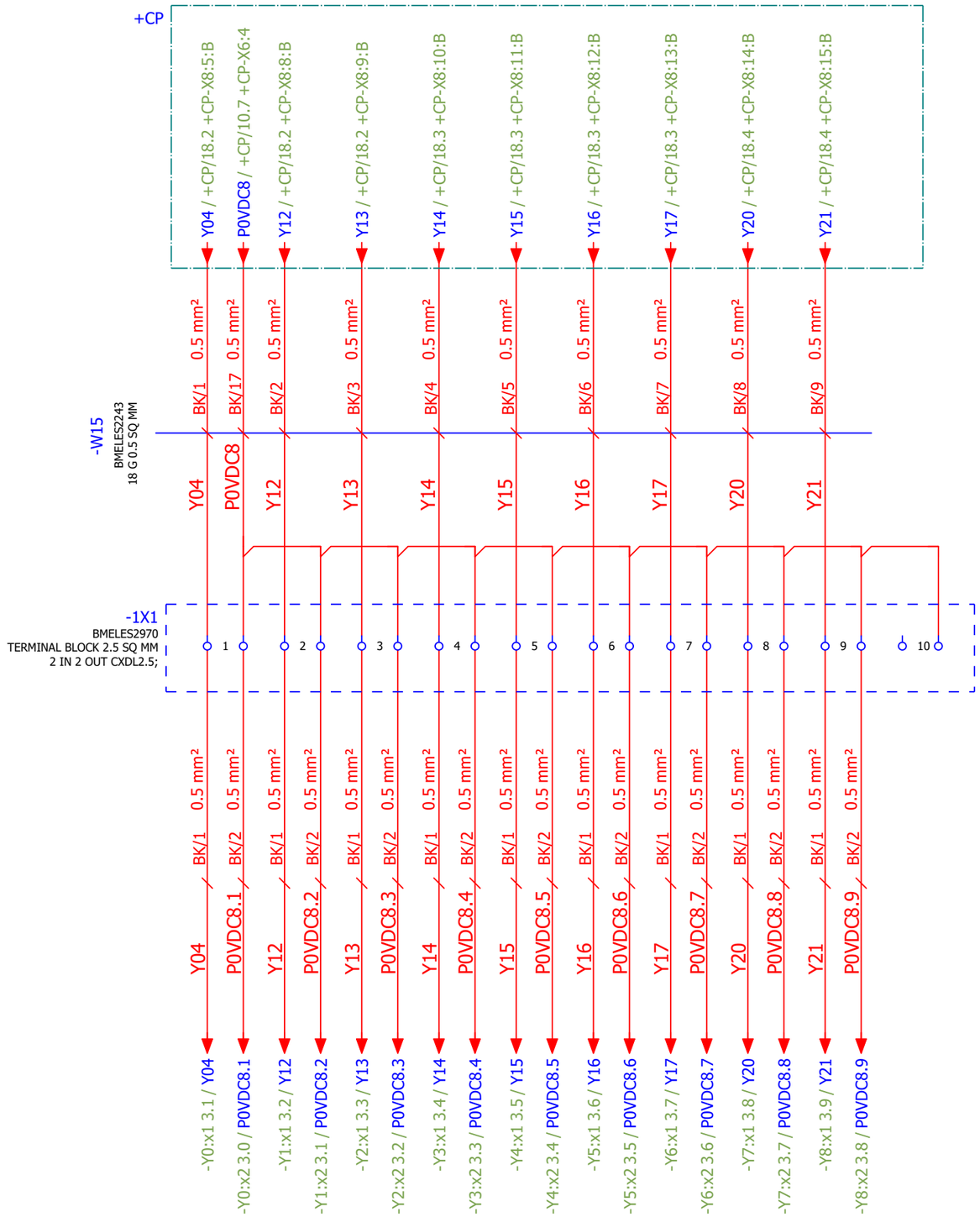
REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	24 PIN I/O INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 25
Modification	Date	Name	Original		Replacement of	Replaced by			Page 26 / 31

CUSTOMIZE PANEL



TAG NO	DESCRIPTION
Y04	SHUT OFF VALVE
Y12	CLAMPING CYL1 FWD SOL
Y13	CLAMPING CYL1 REV SOL
Y14	CLAMPING CYL2 FWD SOL
Y15	CLAMPING CYL2 REV SOL
Y16	SPARE VALVE
Y17	SPARE VALVE
Y20	SPARE VALVE
Y21	SPARE VALVE
-1X1	BMELES2970 TERMINAL BLOCK 2.5 SQ MM 2 IN 2 OUT CXDL2.5;

PNEUMATIC TERMINAL BLOCK



REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	PNEUMATIC TERMINAL BLOCK		=
			Ed.	MANORAJ.SELVARAJ					+ PP
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 2
Modification	Date	Name	Original		Replacement of	Replaced by			Page 28 / 31

REVISION			
Modification	Date	Name	Original
		Ed.	Date
		Appr.	07-09-2023
		MANORAJ.SELVARAJ	

ROBIS

S220 HL HV & LV LIGHT STATION-3

Replacement of

Replaced by

Motherson Automotive & Technology engineering

CYLINDER INTERFACE

FMSPM00856

+ PP

=

Page 29 / 31

3

+FIX/1

SHUT OFF VALVE

Y04
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.1 BK/2 0.5 mm² → P0VDC8.1 / 2.1 -1X1
-Y0 x2 x1
Y04 BK/1 0.5 mm² → Y04 / 2.1 -1X1:1

CLAMP CYL 1 FWD SOL

Y12
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.2 BK/2 0.5 mm² → P0VDC8.2 / 2.1 -1X1
-Y1 x2 x1
Y12 BK/1 0.5 mm² → Y12 / 2.1 -1X1:2

CLAMP CYL 1 REV SOL

Y13
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.3 BK/2 0.5 mm² → P0VDC8.3 / 2.2 -1X1
-Y2 x2 x1
Y13 BK/1 0.5 mm² → Y13 / 2.2 -1X1:3

CLAMP CYL 2 FWD SOL

Y14
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.4 BK/2 0.5 mm² → P0VDC8.4 / 2.2 -1X1
-Y3 x2 x1
Y14 BK/1 0.5 mm² → Y14 / 2.2 -1X1:4

CLAMP CYL 2 REV SOL

Y15
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.5 BK/2 0.5 mm² → P0VDC8.5 / 2.3 -1X1
-Y4 x2 x1
Y15 BK/1 0.5 mm² → Y15 / 2.2 -1X1:5

SPARE VALVE

Y16
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.6 BK/2 0.5 mm² → P0VDC8.6 / 2.3 -1X1
-Y5 x2 x1
Y16 BK/1 0.5 mm² → Y16 / 2.3 -1X1:6

SPARE VALVE

Y17
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.7 BK/2 0.5 mm² → P0VDC8.7 / 2.3 -1X1
-Y6 x2 x1
Y17 BK/1 0.5 mm² → Y17 / 2.3 -1X1:7

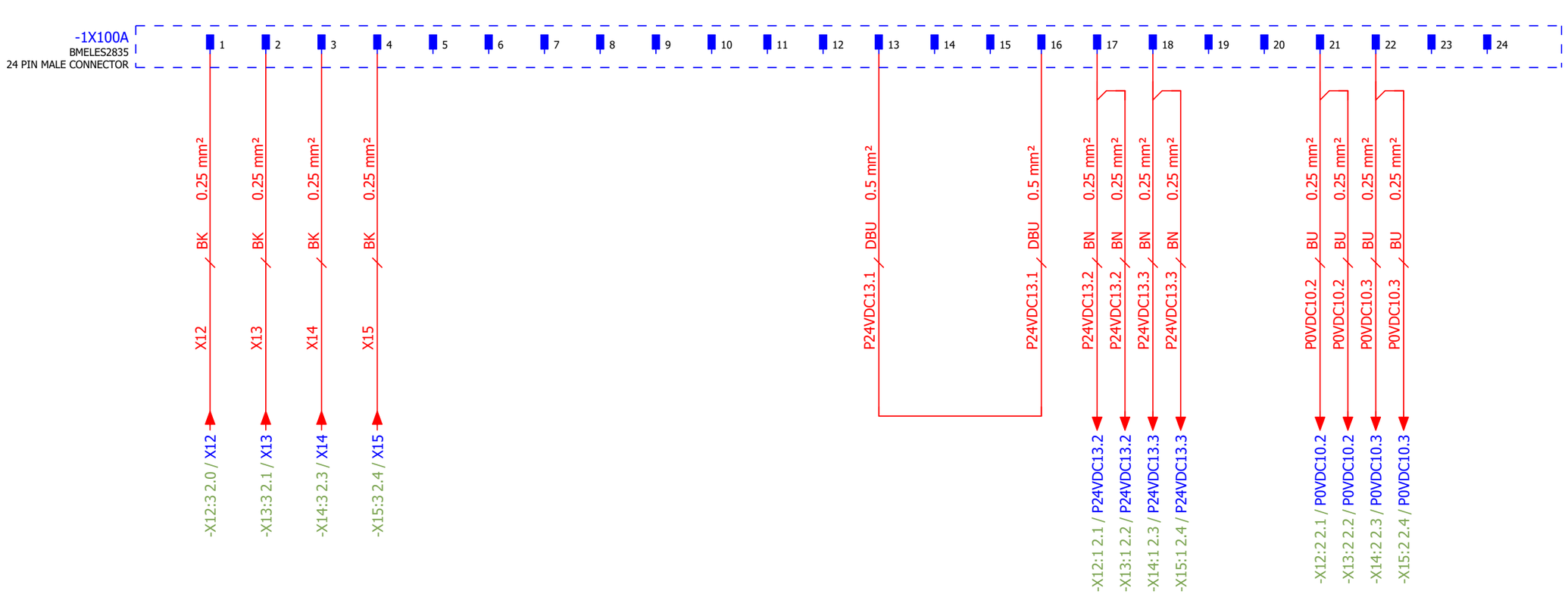
SPARE VALVE

Y20
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.8 BK/2 0.5 mm² → P0VDC8.8 / 2.4 -1X1
-Y7 x2 x1
Y20 BK/1 0.5 mm² → Y20 / 2.3 -1X1:8

SPARE VALVE

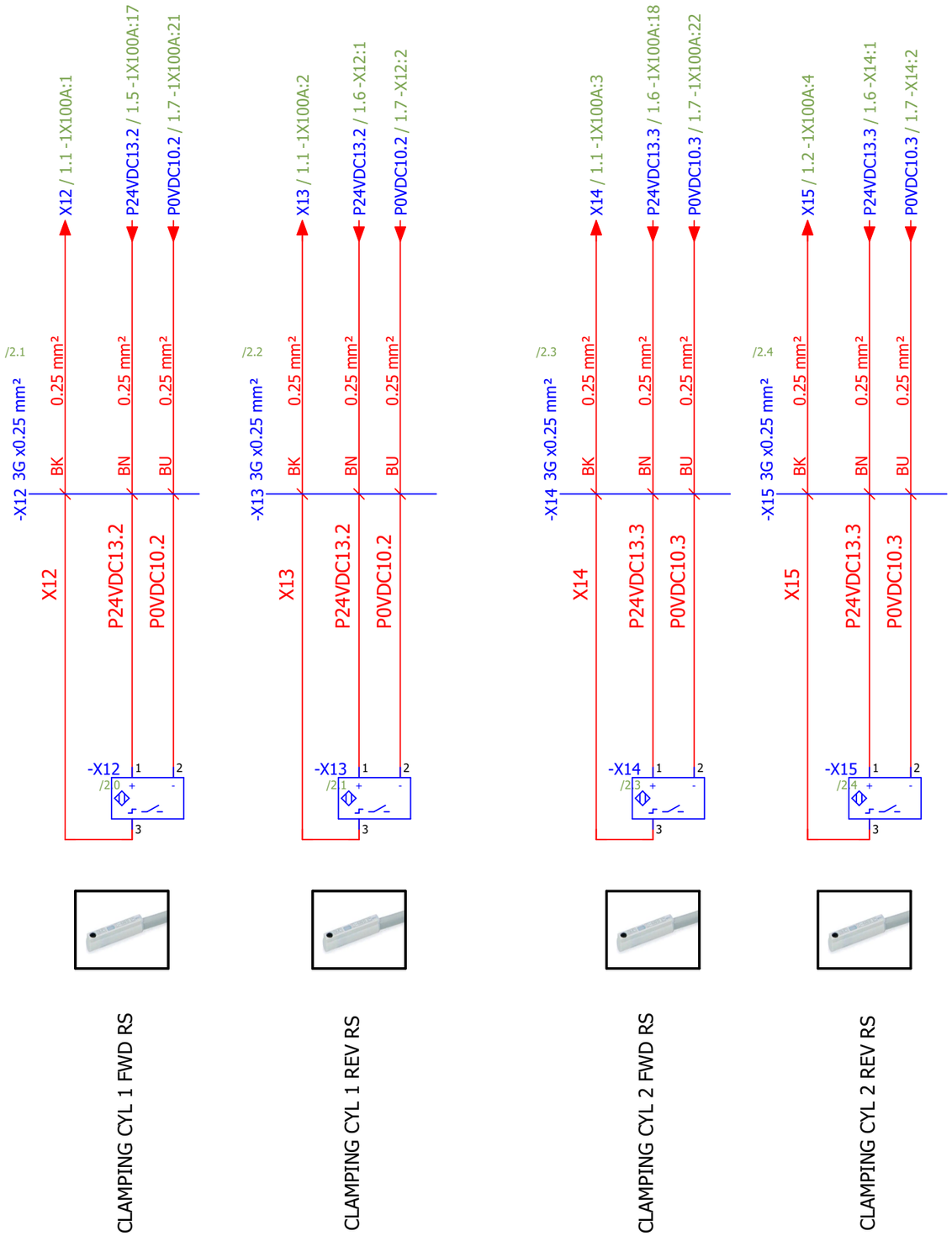
Y21
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.9 BK/2 0.5 mm² → P0VDC8.9 / 2.4 -1X1
-Y8 x2 x1
Y21 BK/1 0.5 mm² → Y21 / 2.4 -1X1:9

24 PIN I/O INTERFACE



REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	24 PIN I/O INTERFACE	=	
			Ed.	MANORAJ.SELVARAJ				+ FIX	
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 1
Modification	Date	Name	Original		Replacement of	Replaced by			Page 30 / 31

REED SWITCH INTERFACE



1

REVISION			Date	31-07-2023	ROBIS	Motherson Automotive & Technology engineering	REED SWITCH INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ FIX
			Appr.		S220 HL HV & LV LIGHT STATION-3			FMSPM00856	Page 2
Modification	Date	Name	Original		Replacement of	Replaced by			Page 31 / 31